

The basic rules of combinatorics

Two rules decide every counting problem: “and” multiplies, “or” adds.

LESSON 155–157

3 × 40 MIN

MATEMATIKA 7, §65

S8 13

LESSON CLOCK

15'

30'

30'

30'

15'

Counting without listing	15 min
The multiplication rule	30 min
The addition rule	30 min
Trees and tables	30 min
With or without repetition	15 min

LEARNING OBJECTIVES

- State and apply the multiplication rule and the addition rule.
- Decide from the wording which rule a problem needs.
- Draw a tree diagram and a table of outcomes.
- Count outcomes with and without repetition.

TEXTBOOK

National	pp. 462–472
Cambridge	Stage 8, pp. 126–134
Workbook	Exercise 13.1

01

Explanation

Counting without listing

How many two-course meals can be made from 3 soups and 4 main dishes? The list has 12 lines; the answer can be had without writing any of them.

PROBLEM	BY LISTING	BY RULE
3 soups, 4 mains	12 lines	$3 \cdot 4$
10 digits, 4-figure code	10 000 lines	10^4
26 letters, 3-letter word	17 576 lines	26^3

WHY THE RULES MATTER

Listing works for a dozen outcomes and fails for a thousand. Combinatorics is the art of counting a set without writing it down — which is also what makes probability possible.

The multiplication rule

If a first choice can be made in m ways and, after it, a second in n ways, then the pair of choices can be made in $m \cdot n$ ways.

“and” $\implies m \cdot n$

PROBLEM	WORKING	ANSWER
3 shirts and 4 ties	$3 \cdot 4$	12
5 starters, 3 mains, 2 puddings	$5 \cdot 3 \cdot 2$	30
a coin and a die	$2 \cdot 6$	12
two dice	$6 \cdot 6$	36

THE RULE EXTENDS TO ANY NUMBER OF STAGES

Three choices give $m \cdot n \cdot p$, four give $m \cdot n \cdot p \cdot q$. Each new stage multiplies, which is why counting problems grow so quickly.

The addition rule

If one thing can be chosen in m ways and another, incompatible with it, in n ways, then one **or** the other can be chosen in $m + n$ ways.

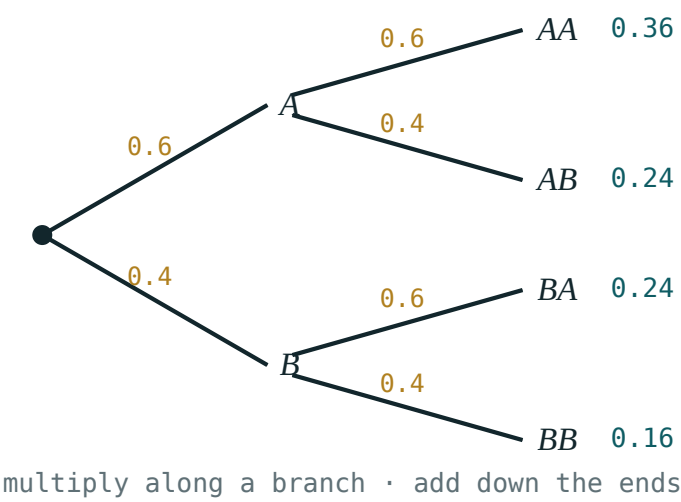
“or” $\implies m + n$

PROBLEM	WORKING	ANSWER
one book from 7 novels or 5 poetry books	$7 + 5$	12
travel by one of 3 buses or 2 trains	$3 + 2$	5
a shirt and a tie, or a jumper alone (3, 4, 6)	$3 \cdot 4 + 6$	18

THE ADDITION RULE NEEDS THE CHOICES TO EXCLUDE EACH OTHER

If 2 of the 7 novels are also counted among the poetry books, adding gives 12 where the true count is 10. Check that nothing is counted twice before adding.

Trees and tables



A tree for two stages of choice

A tree diagram shows every outcome as a path; the number of paths is the product of the branchings — the multiplication rule made visible.

TOOL	GOOD FOR		READS AS			
tree diagram	two or three stages		multiply along a path			
table of outcomes	exactly two stages		rows times columns			
the rules alone	many stages, or large numbers		a single product			

COIN \ DIE	1	2	3	4	5	6
heads	H1	H2	H3	H4	H5	H6
tails	T1	T2	T3	T4	T5	T6

With or without repetition

The second question to ask, after “and or or?”, is whether an item may be used twice.

PROBLEM	REPETITION	WORKING	ANSWER
3-figure codes from 10 digits	allowed	$10 \cdot 10 \cdot 10$	1000
3-figure codes, digits all different	not allowed	$10 \cdot 9 \cdot 8$	720
seating 4 pupils in 4 chairs	not allowed	$4 \cdot 3 \cdot 2 \cdot 1$	24
2-letter codes from 5 letters	allowed	$5 \cdot 5$	25

WITHOUT REPETITION THE NUMBERS COUNT DOWN

10, 9, 8 — each stage has one fewer choice because one item has been used. That descending product is the whole difference between the two cases.

02 Worked examples

EXAMPLE 1 A canteen offers 5 starters, 3 main dishes and 2 puddings. How many three-course meals are possible?

① The three choices are made together — “and”.

② $5 \cdot 3 \cdot 2$

③ $= 30$

ANSWER

30

EXAMPLE 2 How many 3-figure codes can be made from 10 digits if the digits must all differ?

① First digit: 10 ways.

② Second: 9, since one is used.

③ Third: 8.

④ $10 \cdot 9 \cdot 8 = 720$

ANSWER

720

EXAMPLE**3**

A pupil may take one of 7 novels or one of 5 poetry books, with no book in both lists. How many choices?

① The two options exclude each other — “or”.

② $7 + 5$

③ $= 12$

ANSWER**12****03 Interactive model**

Have the class list all 12 shirt-and-tie outfits by hand, then ask for the number with 8 shirts and 9 ties; the rule is welcomed rather than imposed.

And, or, and repeats

3 shirts **and** 4 ties give:

7 novels **or** 5 poetry books give:

A coin and a die give:

Two dice give:

12

24

36

6

3-figure codes from 10 digits, repeats allowed:

30

720

1000

120

The same with all digits different:

720

1000

504

90

Seating 4 pupils in 4 chairs:

4

12

16

24

The addition rule requires the options to be:

equal

mutually exclusive

ordered

repeated

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Combinatorics	Kombinatorika	Комбинаторика
Multiplication rule	Ko'paytirish qoidasi	Правило умножения
Addition rule	Qo'shish qoidasi	Правило сложения
Outcome	Natija	Исход
Tree diagram	Daraxt sxemasi	Дерево вариантов
With repetition	Takrorlanish bilan	С повторением
Without repetition	Takrorlanishsiz	Без повторения
Mutually exclusive	Birgalikda bo'lmagan	Взаимоисключающие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“And” calls for:

adding

multiplying

subtracting

dividing

“Or” calls for:

adding

multiplying

subtracting

dividing

5 starters, 3 mains, 2 puddings give:

10

30

60

15

Without repetition the numbers:

stay the same

count down

count up

double

3-figure codes with different digits number:

720

1000

30

120

A tree diagram shows each outcome as:

a branch

a path

a node

a row

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 3 shirts and 4 ties

ANSWER 12

2. 7 novels or 5 poetry books

ANSWER 12

3. A coin and a die

ANSWER 12

4. Two dice

ANSWER 36

5. Two coins

ANSWER 4

6. 5 starters, 3 mains, 2 puddings

ANSWER 30

7. 3 buses or 2 trains

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. 3-figure codes from 10 digits, repeats allowed

ANSWER 1000

2. The same, all digits different

ANSWER 720

3. Seating 4 pupils in 4 chairs

ANSWER 24

4. 2-letter codes from 5 letters, repeats allowed

ANSWER 25

5. The same without repeats

ANSWER 20

6. A shirt and a tie (3, 4) or a jumper (6)

ANSWER 18

7. Three coins

ANSWER 8

Hard · stretch and reason

7 PROBLEMS

1. Car plates: 3 letters from 26 then 3 digits, repeats allowed

ANSWER $26^3 \cdot 10^3 = 17\,576\,000$

2. How many 4-figure PINs have all digits different?

ANSWER 5040

3. How many 3-digit numbers are there in all?

ANSWER 900

4. How many 3-digit numbers have no repeated digit?

ANSWER 648

5. How many even 3-figure codes from 10 digits, repeats allowed?

ANSWER 500

6. Why does the addition rule fail if the lists overlap?

ANSWER The common items are counted twice

7. Arrangements of the 5 letters of a word with all letters different

ANSWER 120

07 Homework

Homework — 5 tasks

Ask two questions first: “and or or?”, then “may anything repeat?”

1. A shop has 6 shirts and 5 ties. How many outfits?
2. How many 4-figure codes can be made from 10 digits with repeats allowed, and how many without?
3. A pupil chooses one of 8 novels or one of 6 science books. How many choices?
4. Draw a tree diagram for tossing three coins and count the outcomes.

5. In how many ways can 5 pupils be seated in 5 chairs?